Wu et al. 2020 — Acute Myeloid Leukemia

Hematologic · Microwell-seq · 96,875 malignant cells (36/38 samples) · counts (log1p of normalize_total)

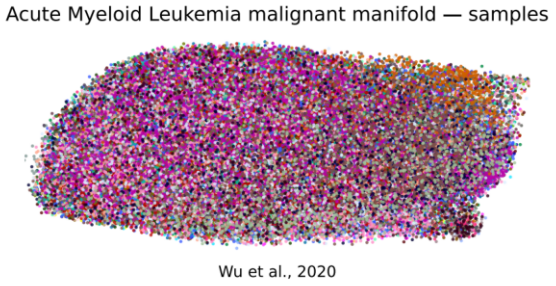
GO / NO-GO GATE LEDGER

Gate 1	**GO**	Mixing + continuity. iLISI 1.00 (fully patient-segregated) → 10.2 of 36 samples after Harmony (θ=10); kNN graph = 1 component. One continuous, uniformly low-cycling dome.
Gate 2	**GO**	Module geography (non-branching). Organized Gast↔pSom cross-repression (r=−0.64) with a single terminal Crest gradient (Crest-high decile at pseudotime 0.86); no diverging arms.
Gate 3	**NA**	Branching-only. Manifold is a fold-free curve (2 tips, 0 forks); no bifurcation to test.

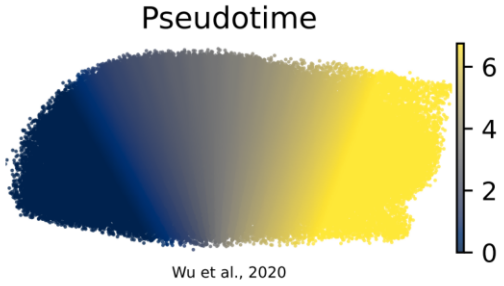
Geometry: Linear (non-branching) · *Crest-arm inversion flag: none (Crest opposes Gast)* · *Hub / r: NA*

STEP FIGURES

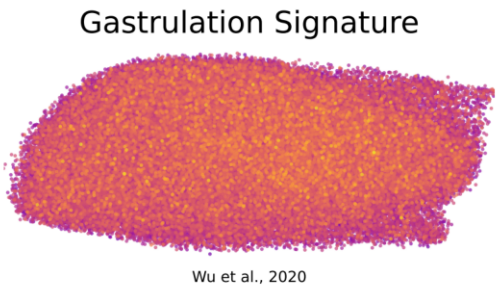
Preprocessing — samples



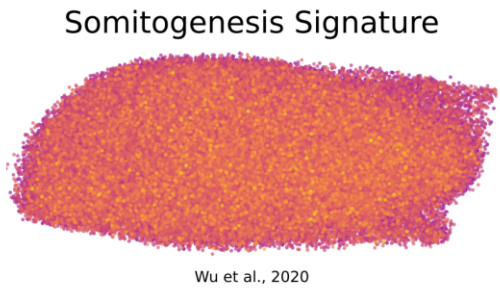
Manifold — pseudotime



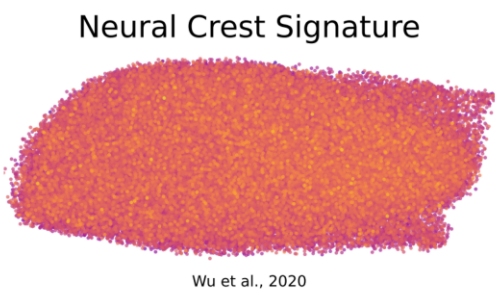
Gastrulation



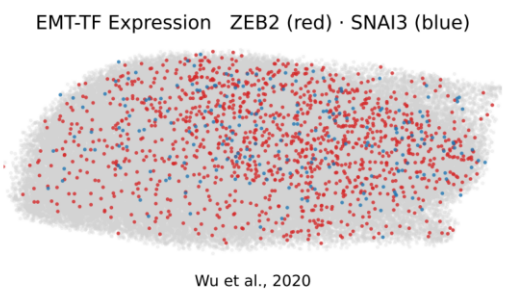
Somitogenesis



Neural Crest



EMT-TF expression



KEY PARAMETER CHOICES

Min-cell threshold	≥100 cells (floor) → drop P18/P13; smooth counts, no dominant multiplicative cliff (top/second 1.35).
Integration	Harmony on sample, θ=10 (iLISI plateau; θ≥20 over-integrates); n_pcs=30, no sc.pp.scale.
Geometry / root	scFates curve, N=20 (fold-free 2 tips/0 forks; PPT tree overfits 5–10 spurious branches).
Root	Gast/pSom-high, Crest-low tip (developmental origin; Crest late/terminal). Cell-cycle-free axis.
EMT-TFs	ZEB2 (5.1%) + SNAI3 (0.9%) — canonical EMT-TF panel largely silent in this myeloid malignancy.

VERDICT: Pattern reproduces (non-branching). Single continuous gastrulation→neural-crest axis with organized Gast↔pSom cross-repression; no bifurcation. G1 GO / G2 GO / G3 NA.